

Natural Disasters and U.S. Food Supply Chains^{*}

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Abstract

This study quantifies how natural disasters affect U.S. food companies and evaluates which diversification strategies mitigate or intensify these effects. We compile a long panel from 1976 to 2023 that links county-level disaster incidence and damages to firm outcomes and supplier–customer relationships in the food sector. The analysis shows that hurricanes are the most consistently economically harmful natural disaster for food processing companies. A firm’s geographical and business structures can mitigate the impact of these shocks. Less customer concentration, more international customer-supplier relationships, and more geographically dispersed branches within the U.S. are all factors associated with smaller effects of hurricanes on aggregate performance. The results are robust to alternative measures of disaster intensity, weighting by firm size, and excluding the COVID-19 period. Overall, the evidence highlights the vulnerability of food production networks to climate-related shocks and informs firm strategy and policy design aimed at strengthening supply-chain resilience.

^{*}Data used in this article are retrieved from proprietary sources, but they may be available to researchers at some institutions. Replication code files can be found at <https://github.com/Chengwei-F/Natural-Disasters-Food-Supply-Chains>.

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1 Introduction

Natural disasters cause large economic disruptions. When a disaster strikes a region, economic activity in that region is directly affected, and other agents are indirectly affected via interconnected linkages. Firms are embedded in supply-chain networks that connect suppliers and customers across regions, and a localized shock can propagate through these linkages to affect firms well beyond the exposed area (Barrot and Sauvagnat, 2016; Carvalho et al., 2021). For food supply chains, this is of particular concern for consumer welfare. When disruptions hit downstream food processors, distributors, and wholesalers, staple food may be stocked out, or prices may rise (Mahajan and Tomar, 2021; Bao, Sun, and Li, 2023). Research on environmental shocks and food systems has focused primarily on farm-gate production and weather shocks, like heat and drought. Less is known about how more rare but intense disaster events affect the firms further downstream. This paper studies how natural disasters affect firms in the U.S. food supply chain and the organizational strategies that mitigate these effects.

Food supply chains have proven vulnerable to a range of disruptions in recent years, and policy interventions to increase resilience have become a priority (Hobbs and Hadachek, 2024). The COVID-19 pandemic exposed fragilities in processing and distribution networks, leading to widespread shortages of meat, dairy, and fresh produce (Hobbs, 2021; Wahdat and Lusk, 2023; Manfredo et al., 2025). Food safety outbreaks have forced recalls that ripple across supplier-customer relationships, and trade disruptions from tariffs or geopolitical tensions have altered sourcing patterns and input costs (Morgan et al., 2022; Spalding et al., 2023). In response, the U.S. Department of Agriculture has implemented strategies to diversify processing capacity and increase industry competition to build supply chain resilience. Identifying vulnerabilities and understanding how firms absorb discrete shocks is central to designing policy that improves resilience without imposing inefficiency costs during normal operating periods.

In this paper, we focus on natural disasters and study how they affect firms in the U.S. food processing, distribution, and wholesale sectors. Our data combine a proprietary dataset of firm transaction records spanning 1976–2023 with county-level disaster exposure from the

Spatial Hazard Events and Losses Database for the United States (SHELDUS). Our design exploits variation in disaster exposure to measure changes in sales growth within supplier-customer pairs. Our preferred empirical specification controls for supplier-customer pair fixed effects, which absorb time-invariant relationship characteristics (e.g., industry overlap, geographical proximity), and region-by-year fixed effects to control for macroeconomic and region-specific conditions. We also control for a set of time-varying firm-specific controls, such as firm size and asset levels.

We first investigate how the most common natural disaster types in the U.S. (hurricanes, flooding, tornadoes, earthquakes, tsunamis, thunderstorms, and extreme heat) affect supplier-customer sales growth. The consequences of disaster shocks also may depend on how firms organize their commercial relationships and operating networks ([Barrot and Sauvagnat, 2016](#)). Once we establish the baseline effects and which disasters are the most consequential, we turn to organizational heterogeneity. Specifically, we examine four means of diversification: customer concentration, business diversification, international diversification, and branch-level geographical dispersion. We interact these indices with one type of disaster exposure, specifically hurricanes, and measure how these business characteristics influence the baseline treatment effect. Furthermore, a disruption to a specific supplier-customer relationship is conceptually different from a disruption to a headquarters or a branch, and the scope for adjustment depends on whether firms can redirect transactions across buyers, shift activity across markets, or reallocate capacity across locations ([Carvalho et al., 2021](#)). We empirically investigate the implications of this distinction by separately estimating the impacts of disasters at the headquarters versus the branch locations.

Our first result is that hurricanes pose an economically and statistically significant threat to firms in the U.S. food supply chain. We find that hurricane exposure, on average, reduces sales growth by about 10 to 20 percentage points, depending on the specification. Each hurricane also reduces the number of food industry establishments operating in a county the next year by an estimated 0.15, and food industry employment by about 6 employees. Other disaster types also carry negative impacts in some specifications, but either due to the limited frequency and statistical power of some disaster types (e.g., tsunamis) or limited

severity (e.g., thunderstorms), these estimates tend to be noisier or inconsistent across outcomes.

Having established the baseline effects of the disaster types, we use hurricane occurrences to study how business organizational strategies mitigate the negative effects. In the moderation analysis, we first find that firms with a more concentrated customer base experience more severe negative impacts from hurricanes. Second, we find some evidence that firms that are diversified internationally mitigate the effects. Third, using several measures of the geographical distribution of branches, we find that firms with more branches and more clusters of branches also mitigate the effects of hurricanes. In general, these results support the notion that diversification, in different forms, can be one way to build resilience to relevant environmental shocks in food supply chains.

A central contribution of this work is to measure risks in food production and their implications for welfare and policymaking. Prior work measures the impacts of climate change on farm-level agricultural outcomes and the threat that future heat and drought pose to future food supply and social welfare ([Deschênes and Greenstone, 2007](#); [Schlenker and Roberts, 2009](#); [Hultgren et al., 2025](#)). However, less is known about how environmental shocks threaten food supply and prices beyond the farm gate. Some past studies document that food prices increase temporarily after a tropical storm or hurricane ([Gagnon and López-Salido, 2020](#); [Bao, Sun, and Li, 2023](#)). [Yim and Dall’Erba \(2025\)](#) measure how drought ripples from farm production to state agricultural exports and food manufacturing. Our findings add to this literature by showing to what extent large, infrastructure-damaging events materialize in economic outcomes in the middle of the food supply chain.

We also contribute to a strand of literature measuring the propagation of shocks through production networks and how business organizations influence how they manifest. Theoretical work shows that firm-level shocks can generate aggregate fluctuations when firms are connected through input-output linkages and supplier–customer networks ([Acemoglu et al., 2012](#); [Acemoglu and Tahbaz-Salehi, 2025](#)). Empirical studies similarly show that disruptions can transmit along supply-chain links and affect firms that are not directly hit by the original

shock (Barrot and Sauvagnat, 2016; Boehm, Flaaen, and Pandalai-Nayar, 2019; Carvalho et al., 2021). Recent evidence on climate exposure in global supply-chain networks further shows that physical climate shocks at supplier locations reduce both supplier and customer operating income and induce customer adjustment of supplier relationships (Pankratz and Schiller, 2024).

Within agriculture and food production, several studies use recent shocks to understand how business organization affects resilience. Stevens and Teal (2024) find that horizontal diversification, specifically, improved firm performance throughout the COVID-19 pandemic. Others have studied how market structure and competition influence outcomes amidst shocks. Hadachek, Ma, and Sexton (2024) provide simulation-based evidence that additional processing capacity only meaningfully increases resilience if it stimulates competition. Richards, Polyviou, and Ubilava (2025) provide empirical evidence that more concentrated markets were actually more resilient to a major supply shock. We provide new insights to this literature by showing that diversification in U.S. food supply chains is one effective way to mitigate the economic losses from natural disasters.

2 Data and Descriptive Statistics

We merge panel data from multiple sources on supplier-customer transaction sales growth and a record of natural disaster frequency as an external shock to the food supply network. We further analyze how sales concentration, business diversification, and international geographical diversification interact with the presence of natural disasters to determine how these factors may moderate or exacerbate the impact of the shocks. Additionally, to enhance our analysis, we leverage establishment- and branch-level data to examine how the geographical clustering of food processing branches contributes to the aggregation of shocks to the supply chain. Appendix Table A1 displays the summary statistics of the main variables.

2.1 Natural Disasters

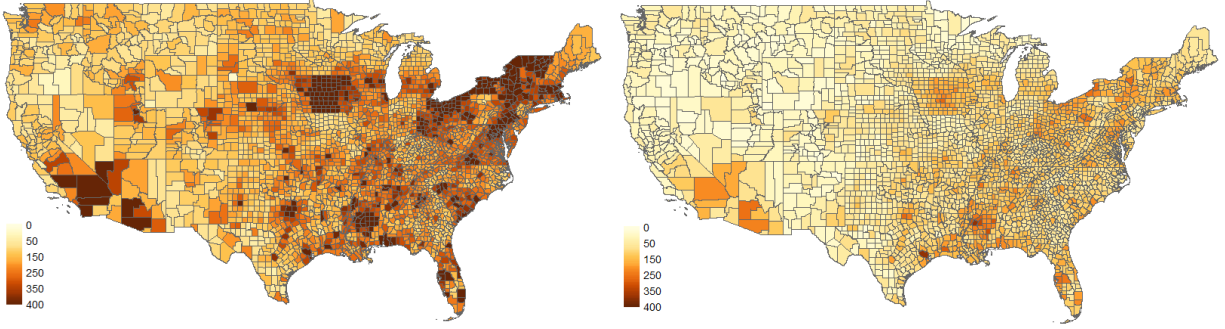
We obtain natural disaster information from the National Oceanic and Atmospheric Administration (NOAA) and the Spatial Hazard Events and Losses Database for the United States

(SHELDUS), maintained by Arizona State University. Our dataset includes a comprehensive (i.e., the universe) record of all 18 classifications of natural disasters from 1976 to 2023 that occur in the U.S.¹, which serve as our primary indicator of disaster frequency. The SHELDUS dataset also includes information about the magnitude of each disaster, measured by estimated property damage. We use this measure of disaster intensity to characterize a secondary measure of disaster exposure as the count of *large* disasters, defined as those causing property damage exceeding \$1 million (adjusted to 2023 dollars).² This characterization of large disasters mirrors past classifications of isolating disasters that plausibly cause disruption (Barrot and Sauvagnat, 2016). We present estimates using both the frequency of all disasters and the frequency of large disasters as measures of disaster exposure in the results section.

Figure 1 compares the frequency of all 18 types of natural disasters with the subset of disasters that resulted in property damage across the United States. The left panel (a) illustrates the widespread distribution of natural disasters, with significant occurrences across regions such as the central U.S., southeastern states, and parts of the western U.S. In contrast, the right panel (b) focuses on our measure of large disasters, highlighting specific areas where natural disasters have caused more than \$1 million in property damage, measured in 2023 U.S. dollars. The southwestern United States, particularly Southern California and Arizona, shows a high concentration of damaging events, driven by wildfires, earthquakes, and severe storms. The southeastern United States, including Florida and the Gulf Coast, is also prominently affected, reflecting the frequent impact of hurricanes and tropical storms. The central U.S., particularly along the Mississippi River and in Iowa, demonstrates a notable frequency of large disasters, often related to severe flooding and storms. Additionally, the northeastern United States, including areas like New York and New Jersey, shows a concentration of property-damaging events, typically associated with severe storms and flooding.

¹The 18 types of natural disasters are: Avalanche, Coastal, Drought, Earthquake, Flood, Fog, Hail, Heat, Hurricane/Tropical Storm, Landslide, Lightning, Severe Thunderstorm, Tornado, Tsunami/Seiche, Volcano, Wildfire, Wind, and Winter Weather.

²Because our empirical focus is on downstream food firms, a property-damage-based disaster measure may better capture events that are likely to disrupt physical assets, infrastructure, and local operations. Excluding lower-damage events also reduces attenuation bias from including minor or noisy disaster reports with limited economic content.



(a) All Disasters

(b) Large Disasters

Figure 1: Frequency of All Disasters and Large Disasters

Note: The figure maps county-level counts of natural disaster events in the United States over 1976–2023. Panel (a) reports the frequency of all recorded disasters, and panel (b) reports the frequency of damaging disasters, defined as events with property damage exceeding 1 million US dollars. Disaster data are from SHELATUS.

Because damaging disasters are often concentrated in densely populated and infrastructure-intensive areas, it is important to clarify how these locations relate to the food industry. Figure 2 maps the spatial distribution of food-industry establishments across the mainland United States. The figure shows that food-industry establishments are spatially concentrated around major population centers, transportation corridors, and distribution regions, including the Northeast corridor, the Great Lakes region, California, Florida, Texas, and other large metropolitan areas. This pattern suggests that areas with large recorded property damages are also often areas where food-industry establishments and consumer markets are located. Therefore, the overlap between damaging disasters and food-industry activity is relevant for studying downstream food-firm outcomes. At the same time, this spatial pattern highlights important context for our analysis. Because our unit of analysis is food industry firms, the relevant variation in disaster exposure arises from events affecting the areas where these firms are located. Therefore, environmental shocks that impact agricultural production more broadly, like extreme heat and drought, may differ from those that impact downstream firms.

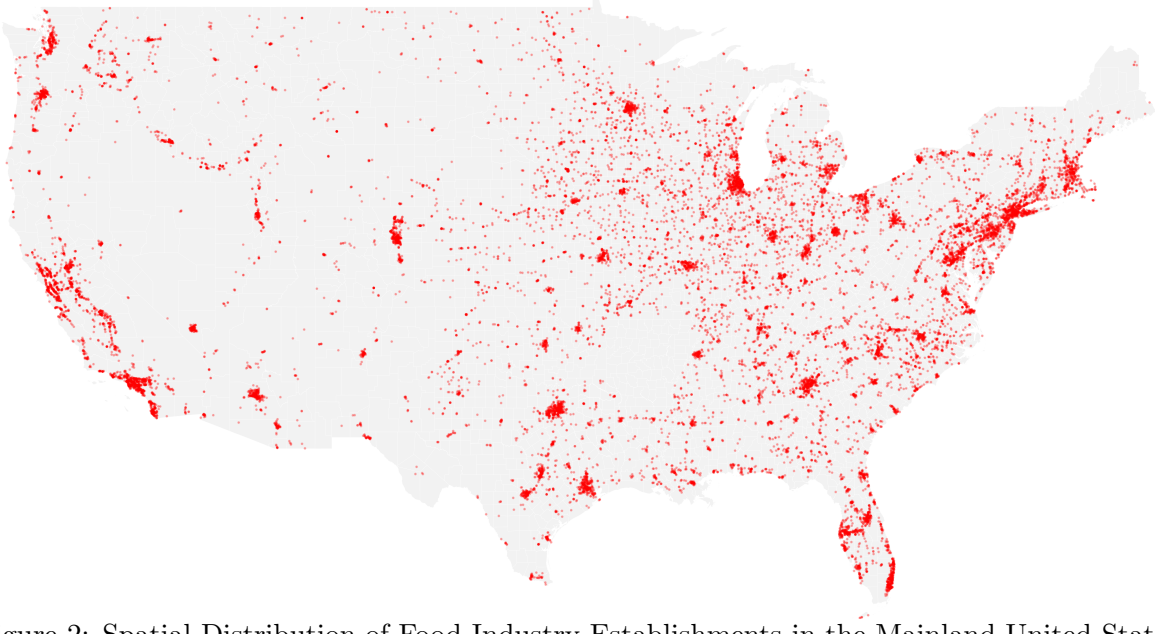


Figure 2: Spatial Distribution of Food-Industry Establishments in the Mainland United States
 Note: The figure maps the spatial distribution of food-industry establishments in the mainland United States using DataAxle Business History data for 2022. Each point represents one establishment.

Figure 3 illustrates the annual trend in the number of natural disasters in the United States from 1976 to 2023. The total number of disasters, represented by black lines and triangles, fluctuates significantly over time, reflecting the unpredictable nature of these events. The colored lines plot the annual property damage from six different types of natural disasters. Hurricane and flooding events (where flooding may be the result of a hurricane) tend to result in the highest amounts of property damage, with Hurricane Katrina in 2005 causing the largest annual property damage amounts. These summary statistics of disaster occurrence and damage are also presented in Appendix Table A2, showing that hurricanes (30.6% of the total property damage) and flooding (31.6%) are the leading natural disasters in terms of property damage.

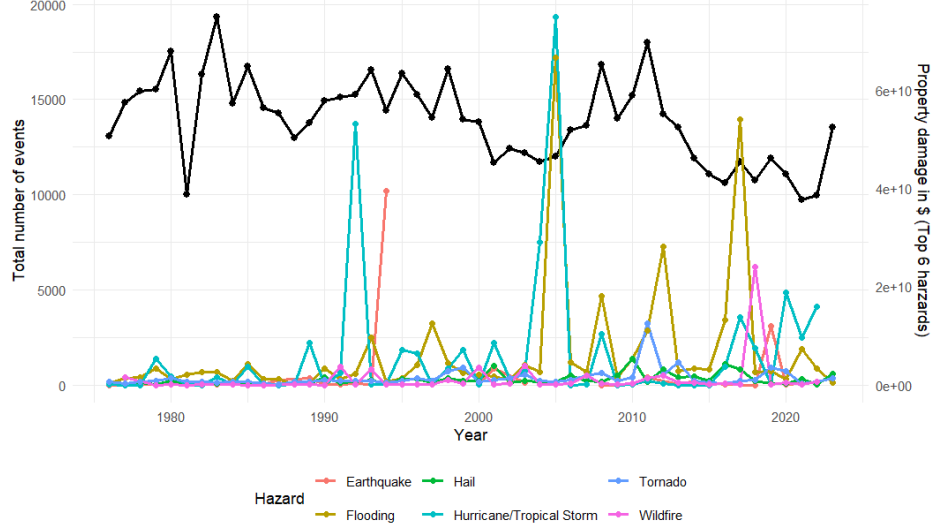


Figure 3: Trends in Natural Disaster Frequency and Property Damage

Note: The figure plots annual disaster frequency and property damage in the United States over 1976–2023. The black line (left axis) reports the total number of disaster events recorded in SHELUDS each year. Colored lines (right axis) report annual property damages, in dollars, for selected hazard types (earthquake, flooding, hail, hurricane/tropical storm, tornado, and wildfire).

2.2 Firm Sales

Supplier-customer pair sales data and company information (including headquarters location³, Return on Assets (ROA), Tobin’s Q, and other relevant variables) are retrieved from the Compustat database available through Wharton Research Data Services. Our sample spans the period from 1976 to 2023 and is restricted to U.S. food-related companies, as defined by the adjusted 4-digit SIC code. For the heterogeneity analysis, we group firms into seven broad food-sector categories: production, meat/dairy, milling/baking, beverage, food wholesale, miscellaneous food, and all other food sectors. These categories are constructed from the SIC-based industry definitions used to build our food-sector sample (Appendix Table A3). We use this grouping to distinguish product types and broad positions in the agri-food supply chain, particularly upstream production-oriented activities versus downstream processing, beverage, and wholesale activities, while maintaining sufficient observations within each category for sector-level estimation. All business activity variables are

³We use whether the food company’s headquarters are hit by a natural disaster to represent an external shock. However, a limitation of Compustat is that it replaces all historical location information with the latest headquarters location. To address this, we manually collected the annual headquarters ZIP codes from the company’s 10-K/Q filings filed with the SEC, allowing us to reconstruct time-varying headquarters locations and more accurately measure disaster exposure. This procedure focuses on key variables that directly affect our identification strategy, and therefore reduces measurement error in disaster exposure.

winsorized at the 99th percentile of their distributions to limit the influence of outliers. The variables Supplier-Customer Sales, Principal Customer Sales, and Segment Sales each capture different facets of our analysis: Supplier-Customer Sales serve as the baseline outcome variable, which captures a comprehensive record of transactions between suppliers and their customers. Principal Customer Sales measures the change in sales concentration, which reflects the share of a supplier’s total sales made to key customers, defined as customers whose purchases account for at least 10%. A higher share of sales to principal customers indicates a less diversified customer base. Segment Sales represent segment sales diversification, reflecting the distribution of sales across different business segments (e.g., sales across business lines) and different international geographical segments.

Supplier-Customer Sales

The initial sample comprises 611 food companies (defined by SIC codes, see Table [A3](#)), including all supplier-to-customer transaction-year observations in the Compustat database, spanning the period from 1976 to 2023. For each supplier-customer pair, we calculate sales growth by comparing sales in the current quarter with those in the corresponding quarter of the previous year. Our data represent nearly all of the reporting companies’ total sales as reported in their 10-K/Q filings. Each customer within the same company is uniquely identified by a customer identification number. This comprehensive coverage ensures that our sample accurately reflects the reporting companies.

2.3 Business Characteristics

We construct four firm-related characteristics used in the analysis: Change in sales concentration, business sales diversification, international sales diversification, and geographical organization of branch networks. These variables capture different dimensions of how firms structure their sales base and operational footprint, which may shape their exposure to disaster-related disruptions.

Change in Sales Concentration

To understand customer concentration and measure the change in sales concentration of a company, we use Compustat supplier-customer data combined with customer information disclosed in SEC filings under Statement of Financial Accounting Standards (SFAS) No. 131. This reporting rule requires firms to disclose major customers accounting for at least 10% of total sales. Using these disclosures, we construct a change in sales concentration measure following [Autor et al. \(2020\)](#), where a higher value indicates a more concentrated customer base and therefore lower customer diversification. This measure captures changes over time in firms' dependence on key customers, which is one potential margin through which disaster shocks may be amplified or absorbed. Appendix Figure [A1](#) shows that average sales concentration remains relatively stable over time, with declining cross-firm dispersion.

Business Segment Sales

We measure business and international diversification using Compustat segment data. Business diversification is based on the distribution of sales across business segments, while international diversification is based on the distribution of sales across geographical segments. Using these data, we construct diversification indices that summarize how broadly a firm's sales are distributed across products and international markets. These measures allow us to distinguish diversification across business lines from diversification across geographical markets, which may have different implications for resilience. Appendix Figure [A2](#) indicates a long-run decline in business diversification and a gradual increase in international diversification, consistent with greater focus on core activities alongside increased global market exposure.

Establishment and Branch Data

To capture firms' operational geography beyond headquarters location, we use two additional data sources. First, we use the Tapestry dataset to obtain county-level information on food-industry establishments and employment, which allows us to examine local industry adjustments following disasters. Second, we use DataAxle Business History data to identify branch locations and parent-subsidiary links, which allows us to measure the spatial organization of firm operations. DataAxle Business History provides establishment-level information on business locations, corporate hierarchy, employment, sales, and other char-

acteristics. A feature of this dataset is that it focuses on firms whose branches are solely located in the U.S. In 2023, this branch-level analysis includes 3,349 parent companies and 24,327 branches, providing a large domestic network for examining how U.S. disasters affect branch exposure and aggregate firm performance. Based on the branch-location data, we construct three measures of geographical organization: the number of branches, the number of spatial clusters, and the share of branches located in clusters. These variables capture the extent to which a firm’s operations are geographically dispersed or concentrated across space.

Figure A3 illustrates two stylized branch-network configurations. In panel (a), branches are more loosely distributed across space. In panel (b), branches are more tightly grouped within clusters, while clusters themselves remain geographically separated. This distinction motivates our cluster-based measures of geographical organization and highlights the trade-off between local concentration and spatial dispersion. Firms with more dispersed operations may be less exposed to localized shocks, whereas clustered operations may facilitate coordination and mutual support within affected areas.

Appendix Figure A4 shows that, on average, firms maintain between 10 and 15 branches, with most branches belonging to one or more clusters. The average number of clusters remains relatively stable over time, suggesting persistent use of geographically organized branch networks.

3 Measurement and Empirical Framework

In this section, we define the measurement of firm diversification and outline the empirical framework used in our analysis. For certain types of diversification, such as the change in sales concentration, measurement is straightforward; we use the firm’s change in sales concentration as an index. However, for more complex dimensions like business and international sales diversification, we employ a sales-based diversification index based on the Herfindahl–Hirschman index (Hirschman, 1980; Herfindahl, 1950). This index provides a comprehensive measure of diversification across different business products and international customers, allowing for a nuanced analysis of how each dimension impacts firm performance.

3.1 Measurement of Diversification

$$D = 100 \times \left(1 - \sum_{i=1}^n \left(\frac{r_i}{S} \right)^2 \right) \quad (1)$$

The diversification measure shown in the equation is commonly used to quantify market concentration and, conversely, to measure diversification. Here r_i represents the sales from segment i , and S is the total sales across all segments, $S = \sum_{i=1}^n r_i$. If the firm's sales were concentrated in a single segment, $\left(\frac{r_i}{S}\right)^2$ for that segment would equal 1, and the diversification index D would equal zero, indicating low diversification. Conversely, if sales were spread evenly across many segments, each $\left(\frac{r_i}{S}\right)^2$ term would be smaller, and the index D would approach 100, indicating high diversification.

We examine customer concentration, business diversification, international diversification, and branch-network geography as potential moderators of disaster impacts. First, customer concentration is measured using the distribution of sales across major customers; lower concentration indicates a broader customer base. Second, business diversification is measured using the distribution of sales across business segments, and international diversification is measured using the distribution of sales across geographical segments. These segment-based measures capture different dimensions of sales dispersion and need not have the same economic implications. Business diversification may reflect product-line breadth or organizational complexity, whereas international diversification captures exposure to multiple geographical markets.

Third, to measure geographical clustering of firm operations, we group branch locations into spatial clusters using a density-based clustering method (HDBSCAN). Intuitively, this approach identifies groups of branches that are geographically close to each other while allowing for irregular cluster shapes and varying density across space. Based on these clusters, we construct three measures of geographical organization: the number of clusters, the number of branches, and the share of branches located within clusters. These measures capture the extent to which a firm's operations are geographically concentrated or dispersed. A higher number of clusters or a more dispersed cluster structure indicates greater geographical di-

versification, which may reduce exposure to localized shocks. A detailed description of the clustering algorithm is provided in the Appendix Section Clustering Methodology (HDB-SCAN).

3.2 Empirical Estimation

In setting up the empirical framework for estimation, we focus on two main aspects: (1) the direct effects of disasters on customer sales and industry outcomes and (2) the moderating influence of diversification strategies on these effects. The following equation outlines the baseline model used in our analysis.

$$\Delta Sale_{i,j,t-4,t} = \alpha_0 + \alpha_1 Frequency_{i,t-4} + \mu_{i,j} + \gamma_t + \epsilon_{i,t} \quad (2)$$

Here, α_1 is the effect of disaster frequency on sales growth, where $\Delta Sale_{i,j,t-4,t}$ is supplier i 's sales growth to customer j between the current quarter and the same quarter in the previous year. $Frequency_{i,t-4}$ is the frequency of natural disasters affecting the county where supplier i is located during the same period as the observed sales growth. $\mu_{i,j}$ and γ_t are supplier-customer pair and year fixed effects, respectively. In more saturated specifications, we include region-by-year fixed effects, and $X_{i,t}$ includes a vector of company financial metrics as control variables, which may be important omitted variables. We cluster standard errors at the county level to account for potential correlation across observations within a county.

To study the effects on other industry outcomes, we make two modifications to the baseline model. First, to obtain a broader view of the impact on the food industry, we regress annual differences in county-level food industry establishments and employment on hurricane exposure, controlling for county and year fixed effects. Second, we measure hurricane exposure as the share of branches affected by a hurricane, rather than disaster frequency in the headquarters county. However, since we do not observe supplier-customer sales at the branch level, the outcome variable reflects the company's aggregate sales growth across all customers and branches. In these specifications, we control for company and year fixed

effects. If individual supplier-customer relationships are disrupted or branch impacts can be absorbed by other branches, the economic effects of natural disasters on branches may be smaller than their impact at the headquarters.

We also explore sector-specific effects using the baseline supplier-customer model. The following two equations are used to estimate the sector-specific disaster effects:

$$\Delta Sale_{i,j,t-4,t} = \alpha_0 + \alpha_1(Frequency_{i,t-4}|Sector_i) + \mu_{i,j} + \gamma_t + \epsilon_{i,t} \quad (3)$$

$$\Delta Sale_{i,j,t-4,t} = \alpha_0 + \alpha_1(Frequency_{i,t-4} \times Sector_i) + \mu_{i,j} + \gamma_t + \epsilon_{i,t} \quad (4)$$

where we separate the food-sector sample into seven broad sectors: production, meat/dairy, milling/baking, beverage, food wholesale, miscellaneous food, and all other food sectors. These models allow us to observe how disaster impacts may vary depending on the nature of the food industry.

Because different disaster types may vary in their relevance for downstream food firms, we first estimate effects for all major disaster types before narrowing the diversification analysis to the type that generates the most consistent disruptions. As we show in the results section, this leads us to focus on hurricanes. We estimate the moderating influence of diversification strategies, focusing on hurricane impacts:

$$\begin{aligned} \Delta Sale_{i,j,t} = & \alpha_0 + \alpha_1 Hurricane_{i,t} + \alpha_2 Diversification_{i,t} \\ & + \alpha_3(Hurricane_{i,t} \times Diversification_{i,t}) + \beta X_{i,t} + \mu_{i,j} + \gamma_t + \epsilon_{i,t} \end{aligned} \quad (5)$$

In this model, $Hurricane_{i,t}$ represents the occurrence of hurricanes, while $Diversification_{i,t}$ reflects the firm's diversification level. The interaction term $Hurricane_{i,t} \times Diversification_{i,t}$ assesses the buffering effect of diversification on hurricane impacts. As in the baseline specification, $\mu_{i,j}$ and γ_t denote supplier-customer pair and year fixed effects, respectively.

Identification in all models exploits variation in whether a supplier’s headquarters county is hit by a natural disaster in a given year, compared within supplier–customer pair sales over time. Supplier–customer fixed effects absorb time-invariant differences across trading relationships, including persistent differences in product type, relationship quality, firm location, and long-run regional characteristics. Year fixed effects absorb aggregate shocks common to all firms in a given year, like recessions. In the more saturated specifications, we further include firm-level financial controls capturing firm scale, profitability, and financial structure, including the number of employees, gross profit margin, ROA, and operating leverage. These controls absorb time-varying factors that may affect sales growth, including local economic shocks unique to the firm. Standard errors are clustered at the county level to allow for correlated shocks among firms exposed to the same local disaster environment.

The identifying assumption is that, conditional on supplier–customer fixed effects, year fixed effects, and time-varying firm controls, natural disasters are not correlated with other unobserved shocks that would otherwise differentially affect supplier–customer sales growth. This assumption is likely to hold for unanticipated events with unpredictable severity, like hurricanes. Nevertheless, even if firms are not directly affected, natural disasters may indirectly contribute to a firm’s sales growth if other interconnected sectors or infrastructure are damaged. We therefore interpret the estimates of disaster-related disruptions to supplier–customer transactions as the gross effect of all channels that may influence sales growth.

4 Results

We divide our analysis into three parts, each addressing different aspects of how disasters affect food companies’ sales growth. First, we examine the impact of disasters on supplier–customer transaction growth when the company’s headquarters is directly affected. This part of the analysis focuses on how disruptions at the headquarters, caused by different types of natural disasters, influence the transaction volume between suppliers and customers. The second part of the analysis shifts to the branch level, where we explore how the percentage of branches affected by different disasters influences the sales growth of the entire company. Finally, we use hurricanes as a specific example – given their frequency and impact – to

investigate how diversification strategies can help mitigate the negative effects. By focusing on both sales diversification and branch geographical dispersion diversification, we find that these strategies play a crucial role in moderating the adverse impact of hurricanes.

Natural Disasters and Sales Growth

We first explore the impact of damaging disasters on the annual transaction sales growth between suppliers and customers, with a specific focus on cases where the headquarters of the supplier is located in a disaster-affected county.

Figure 4 plots the coefficients and 95% confidence intervals for the treatment effects of natural disasters on sales growth. Panel (a) measures disaster exposure as the frequency of all events. Panel (b) only measures the frequency of natural disasters that caused over \$1 million in property damage. Both panels suggest that hurricanes have the strongest and most statistically robust negative effect on food-company supplier–customer sales growth, indicating substantial disruption to transaction flows within the food supply chain. Intuitively, large hurricanes cause a larger decline in sales growth than the average effect of all hurricanes. By contrast, the estimated effects of flooding and tornadoes are smaller in magnitude and not statistically distinguishable from zero in the baseline specification. One reason for the imprecision is that effects may be harder to detect with our empirical design because such disasters can be highly localized, may operate through shorter disruption windows, or may affect specific facilities that are not fully captured by linking headquarters locations to county-level natural disasters. Earthquakes and tsunamis also show negative point estimates, but the wide confidence intervals indicate substantial estimation uncertainty, perhaps due to their limited historical frequency in counties where U.S. food firms are located.⁴

⁴As a point of comparison, Appendix Table A4 examines the effect on all non-food firms’ sales growth in response to the same shocks. The coefficients are qualitatively similar, with hurricanes inducing a 7 percentage point decline, but the effect is not statistically significant. The imprecise estimates may be due to different responses and exposure across a broader set of industries.

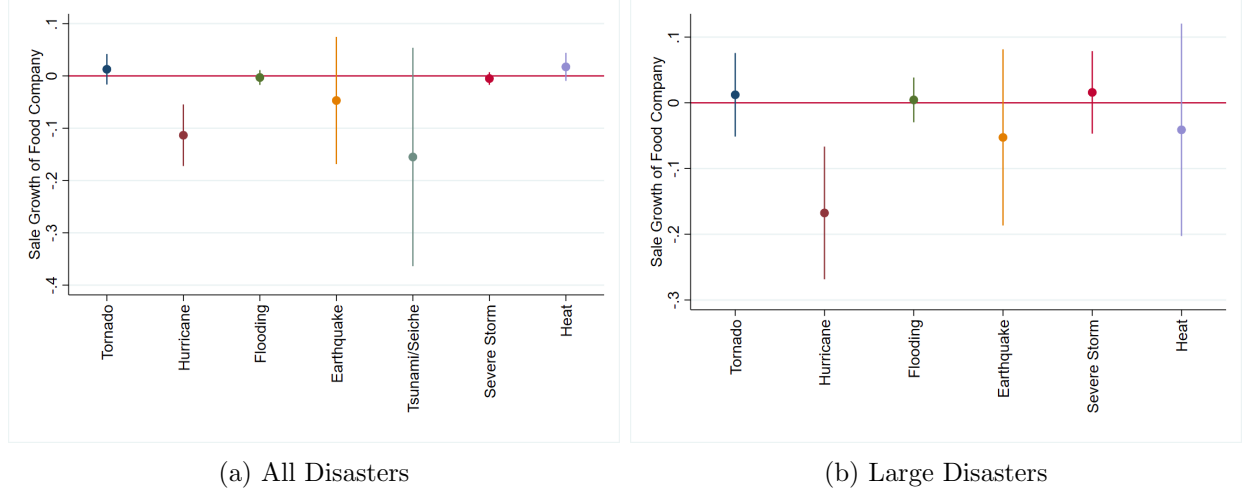


Figure 4: Impact of All and Large Disasters on Sales Growth

Note: The figures plot estimated coefficients from regressions of food-company supplier-customer transaction-level sales growth on exposure to any size of natural disaster (panel a) and disasters that impose over \$1 million in property damage (panel b). The regressions include supplier-customer pair fixed effects and region-by-year fixed effects. Points show the coefficient estimates and vertical bars indicate 95% confidence intervals. Standard errors are clustered at the county level.

Figure 5 illustrates the varying effects of different disasters on different food sectors when the company's headquarters is directly impacted. Blue indicates a positive effect on sales growth, while red shows a negative effect. The left panel shows the full range of coefficients, capturing the effects of all disasters across all food sectors. In the right panel, only statistically significant effects at the 5% level are shown. Damages appear more concentrated in milling and baking, food wholesale, and beverage activities than in some other food-industry sectors. As shown in Figure 2, the firms in our data are more spatially concentrated around major population centers, transportation corridors, and distribution hubs. The observed sector-specific treatment effects may therefore reflect both differences in sector-specific rates of exposure that may contribute to a more precisely estimated treatment effect and sector-specific vulnerabilities to certain types of shocks. We thus treat this sectoral pattern as descriptive evidence, rather than strong evidence that certain industries are by nature more vulnerable to these shocks than others.

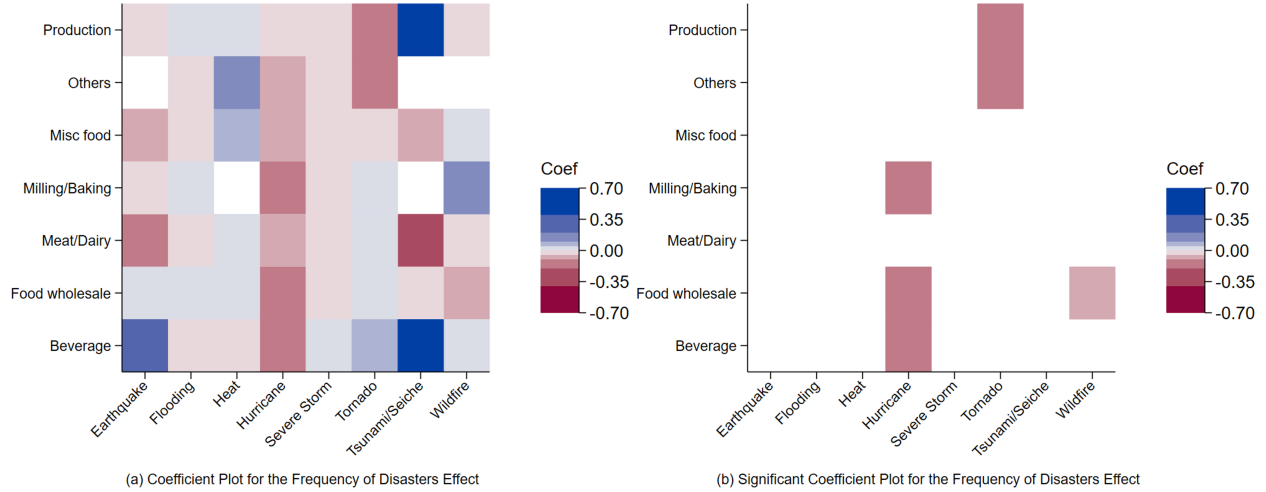


Figure 5: Estimated Effects of Disasters on Sales Growth across Food Sectors: Headquarters Exposure

Note: The figure summarizes estimated disaster effects on supplier-customer transaction sales growth by sector when disaster exposure is measured at the supplier's headquarters county. Panel (a) reports coefficient estimates for each disaster type across food sectors. Panel (b) reports only coefficients that are statistically significant at the 5% level. Colors encode the sign and magnitude of estimates, ranging from blue for positive effects to red for negative effects.

The sector-level patterns also help clarify the likely channels of disruption. Production-oriented activities may be more directly linked to upstream agricultural or input-related shocks, whereas processing, beverage, wholesale, and other downstream activities are more exposed to disruptions in facilities, storage, transportation, and distribution. This distinction is important because the paper focuses on downstream food firms beyond the farm gate, where infrastructure and operational disruptions may matter more than direct agricultural production losses.

Number of Establishments and Employment

One mechanism through which natural disasters impact food companies is through the disruption of infrastructure, which in turn affects the supplier-customer transactions. These disasters can destroy physical establishments, such as factories and warehouses, reducing the company's capacity to operate. Additionally, damaging disasters can lead to population displacement and further destabilization of both production and distribution.

In Panel A of Table 1, we examine the impact of major disasters on changes in food-industry establishment counts. Tsunamis/Seiches have the largest negative and statistically significant effect, while hurricanes are also associated with a statistically significant decline in

establishments. By contrast, the tornado coefficient is positive and statistically significant, but small in magnitude. Given that the outcome is the annual net change in county-level establishment counts, this estimate should be interpreted cautiously: it may reflect business turnover, replacement entry, or the highly localized nature of tornado damage rather than a true expansionary effect on the food industry. Flooding and severe thunderstorms are close to zero, and the earthquake estimate is imprecise, as reflected in its wide confidence interval. Overall, the clearest evidence of establishment losses comes from hurricanes and tsunamis rather than from all disaster types uniformly.

Panel B of Table 1 reports negative employment effects for hurricanes, earthquakes, and tsunamis. Hurricane exposure is associated with a decline of approximately 6 employees, significant at the 10% level. Earthquakes and tsunamis are associated with declines of approximately 146.5 and 56.1 employees, significant at the 1% and 5% levels, respectively. The estimated effects of the remaining disaster types are not statistically significant.

Table 1: Effects of Major Disasters on Changes in Food-Industry Establishment Counts and Employment

Panel A		Difference in Establishment Count					
	Tornado	Hurricane	Flooding	Earthquake	Tsunami	Thunderstorm	Heat
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Disaster	0.0588** (0.029)	-0.1473** (0.061)	0.0063 (0.017)	0.3059 (0.787)	-2.942*** (0.435)	0.0077 (0.011)	-0.1199 (0.080)
County FE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Obs.	64,095	64,095	64,095	64,095	64,095	64,095	64,095
R^2	0.291	0.291	0.291	0.291	0.291	0.291	0.291
Panel B		Difference in Employment Level					
	Tornado	Hurricane	Flooding	Earthquake	Tsunami	Thunderstorm	Heat
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Disaster	-0.4422 (0.718)	-5.969* (3.490)	1.766 (1.630)	-146.5*** (45.29)	-56.082** (25.02)	0.079 (0.653)	0.758 (4.629)
County FE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Obs.	64,095	64,095	64,095	64,095	64,095	64,095	64,095
R^2	0.064	0.064	0.064	0.064	0.064	0.064	0.064

Notes: The dependent variables in Panels A and B are the year-over-year changes in establishment counts and employment levels, respectively. Columns (1)–(7) use tornado, hurricane, flooding, earthquake, tsunami, severe thunderstorm, and heat as the main independent variables, respectively. Standard errors are clustered at the county level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Taken together, the establishment and employment results provide complementary evidence of contraction in the local economic environment. While this analysis cannot include firm-level controls or measure firm-level diversification, it provides county-level measures of establishment counts and employment, unconditional on whether the firm survives the disaster. We therefore interpret the establishment and employment results as aggregate evidence that the firm-level analysis may understate.

Effects of Diversification

Having established that hurricanes are the primary disaster type that consistently hampers food industry outcomes, we now examine whether different forms of diversification moderate these effects. We organize these results around two distinct levels of dependent variables. Tables 2 and 3 use supplier–customer transaction growth as the dependent variable, so the estimated hurricane effects capture disruptions to specific trading relationships. These transaction-level effects are therefore expected to be larger than aggregate firm-level sales effects. By contrast, Table 4 uses firm-level sales growth and examines whether branch-level geographical organization attenuates the aggregate impact of hurricane exposure. This distinction is important for interpreting economic magnitudes: hurricanes reduce transaction-level sales growth by roughly 10–20 percentage points in the supplier–customer specifications, while the firm-level branch estimates are closer to 2–3 percentage points. The latter are smaller because firms may partially absorb or redistribute localized disruptions across branches.

Change in Sales Concentration

Table 2 presents the results of an analysis on the moderation effect of the change in sales concentration on the impact of hurricanes on sales growth. The key interaction term of interest is Hurricane * Δ Sales Concentration, which captures how sales concentration influences the relationship between hurricanes and sales growth.

Table 2: The Moderating Effect of Changes in Sales Concentration

	Sales Growth (t-4, t)				
Panel A: Main Result	(1)	(2)	(3)	(4)	(5)
Hurricane	-0.106*** (0.029)	-0.105** (0.039)	-0.087*** (0.031)	-0.146*** (0.042)	-0.212*** (0.060)
Δ Sales Concentration		0.0006** (3e-4)	0.0006** (3e-4)	0.003*** (0.001)	0.004*** (0.001)
Hurricane * Δ Sales Concentration			-0.0005*** (1e-4)	-0.0014*** (3e-4)	-0.0014*** (3e-4)
Supplier-Customer FE	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES
Financial Controls (Size, ROA, etc.)	NO	NO	NO	YES	YES
Region * Year FE	NO	NO	NO	NO	YES
Observations	8,529	3,341	3,341	1,764	1,750
R-Squared	0.254	0.337	0.340	0.435	0.471
Panel B: Marginal Effects					
Baseline Hurricane Effect			-0.0874*** (0.031)	-0.1455*** (0.042)	-0.2123*** (0.060)
Δ Sales Concentration Increase 10%			-0.0925*** (0.031)	-0.1596*** (0.040)	-0.2265*** (0.060)

Notes: The dependent variable in Panel A is supplier-customer transaction sales growth. Columns (1)-(3) progressively introduce the moderating effect of changes in sales concentration, while Columns (4)-(5) add additional controls. Panel B reports marginal effects implied by the moderation specification. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In Panel A, hurricanes have a statistically significant negative impact on supplier–customer transaction growth across all models. In the baseline model, the hurricane coefficient is -10.6 percentage points, which is about 18% of a standard deviation in typical customer–supplier sales growth variation. As additional controls are included, the estimated effect remains negative and economically large, reaching -21.2 percentage points in the fully specified model. These magnitudes should be interpreted as transaction-level disruptions rather than aggregate firm-level sales losses. They are consistent with the broader production-network literature showing that localized disaster shocks can propagate through supplier–customer relationships and generate economically meaningful losses (Barrot and Sauvagnat, 2016; Boehm, Flaaen, and Pandalai-Nayar, 2019; Carvalho et al., 2021). Related evidence from food markets also shows that major supply-chain disruptions can substantially reduce product availability (Mahajan and Tomar, 2021).

Panel B translates these interaction effects into more intuitive economic magnitudes. In the baseline marginal-effect calculation, hurricane exposure reduces supplier–customer transaction growth by 8.74 percentage points. A 10 percentage point increase in sales concentration makes the implied hurricane effect more negative, from -8.74 to -9.25 percentage points. In the more saturated specification, the implied hurricane effect changes from -21.23 to -22.65 percentage points. Higher customer concentration makes the estimated hurricane effect more negative.

These results suggest that customer concentration can create vulnerability during disruptive events. While a more concentrated customer base may be associated with relational or operational efficiencies in normal periods, it also increases dependence on a smaller set of trading relationships. During hurricanes, this dependence appears to amplify transaction-level sales losses. We therefore interpret the concentration measure as an organizational exposure margin, rather than as a purely exogenous strategic choice.

Segment Diversification

Table 3 reports the moderation effects of business diversification and international diversification on the impact of hurricanes on sales growth, with a specific focus on the interaction terms Hurricane * Business Diversification and Hurricane * International Diversification.

Table 3: The Moderating Effects of Business and International Diversification

Sales Growth (t-4, t)							
Panel A: Main Result	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Hurricane	-0.106*** (0.029)	-0.107*** (0.031)	-0.0961** (0.041)	-0.137*** (0.046)	-0.1916*** (0.047)	-0.193*** (0.052)	-0.196*** (0.051)
Business Diversification		-0.0015 (0.001)	-0.0014 (0.001)				
Hurricane * Business Diversification			-0.0008 (0.001)				
International Diversification				0.0017 (0.001)	0.0016 (0.001)	0.0017 (0.001)	0.0017 (0.001)
Hurricane * Int'l Diversification					0.0033* (0.002)	0.0035* (0.002)	0.0033 (0.002)
Supplier-Customer FE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Financial Controls (Size, ROA, etc.)	NO	NO	NO	NO	NO	YES	YES
Region * Year FE	NO	NO	NO	NO	NO	NO	YES
Observations	8,529	7,770	7,770	7,083	7,083	6,871	6,868
R-Squared	0.254	0.259	0.260	0.249	0.250	0.256	0.280
Panel B: Marginal Effects							
Baseline Hurricane Effect			-0.0961** (0.041)		-0.1916*** (0.047)	-0.1926*** (0.052)	-0.1962*** (0.051)
Business Diversification Level Increase 10			-0.1036*** (0.029)				
International Diversification Level Increase 10					-0.1585*** (0.045)	-0.1577*** (0.048)	-0.1633*** (0.046)

Notes: The dependent variable in Panel A is supplier-customer transaction sales growth. Columns (1)-(3) progressively introduce the moderation effect of business diversification. Columns (4)-(6) introduce the moderation effect of international diversification, and Column (7) further includes region-by-year fixed effects. Panel B reports marginal effects implied by the corresponding moderation specifications. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In Panel A, the results show a statistically significant negative effect of hurricanes on sales growth across all reported specifications, with the baseline hurricane impact ranging from -9.6 percentage points to -19.6 percentage points as additional controls are added. The coefficient on Business Diversification is negative and small, but statistically insignificant, suggesting no clear standalone relationship with sales growth. Likewise, the interaction term Hurricane * Business Diversification is not statistically significant, indicating that business diversification does not materially moderate the negative effects of hurricanes on sales growth. By contrast, International Diversification shows a different pattern: the interaction term Hurricane * International Diversification is positive and statistically significant in the later models, indicating that international diversification helps attenuate hurricane-related losses.

Panel B translates the international-diversification interaction into economic terms. In Column (5), the implied hurricane effect decreases in magnitude from -19.16 percentage points

to -15.85 percentage points when international diversification increases by 10 percentage points. This corresponds to an attenuation of about 3.31 percentage points, or roughly 17 percent of the baseline hurricane effect in that specification. The magnitude suggests that international diversification can meaningfully buffer transaction-level losses, although it does not fully offset the negative effect of hurricane exposure.

Overall, these results indicate that while business diversification does not significantly alter the impact of hurricanes, international diversification plays a critical role in moderating hurricane-related disruptions. By diversifying internationally, firms can mitigate the adverse effects of hurricanes on sales growth, enhancing resilience in disaster-prone environments. This finding highlights the strategic importance of geographical diversification for companies seeking to maintain stability in the face of natural disasters.

Geographical Diversification

Unlike Tables 2 and 3, the geographical diversification analysis uses firm-level sales growth rather than supplier–customer transaction growth, so the coefficients capture aggregate performance responses rather than losses on individual trading relationships. Table 4 presents the moderation effect of different geographical cluster measures on the impact of hurricanes on sales growth, analyzing how the number of clusters, branches, and the percentage of clustered branches influence the relationship between hurricane exposure and sales performance. This analysis demonstrates how the distribution and clustering of branches can serve as a buffer against the concentrated negative impacts of hurricanes.

Table 4: The Moderating Effects of Branch Network Structure

Sales Growth (t-4, t)											
Panel A: Main Result		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Hurricane		-0.021 (0.013)	-0.0204 (0.013)	-0.0296** (0.014)	-0.0299** (0.014)	-0.0204 (0.013)	-0.0296** (0.014)	-0.0299** (0.014)	-0.0202 (0.014)	-0.0341** (0.015)	-0.0344** (0.015)
Num. Cluster			0.0015** (7e-4)	0.0014* (7e-4)	-0.0004 (6e-4)						
Hurricane *				0.0076*** (0.002)	0.0077*** (0.002)						
Num. Cluster											
Num. Branch						0.0001* (6e-5)	0.0001* (6e-5)	-4e-5 (2e-4)			
Hurricane *							0.0005*** (2e-4)	0.0006*** (2e-4)			
Num. Branch									0.0004*** (1e-4)	0.0004** (1e-4)	0.0004** (2e-4)
Percentage of											
Clustered Branch											
Hurricane * %										0.0008*** (3e-4)	0.0009*** (3e-4)
Clustered Branch											
Firm FE		YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year FE		YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Size Control		NO	NO	NO	YES	NO	NO	YES	NO	NO	YES
Observations		19,540	19,254	19,254	19,254	19,254	19,254	19,254	19,254	19,254	19,254
R-Squared		0.326	0.326	0.326	0.326	0.326	0.326	0.326	0.326	0.326	0.326
Panel B: Marginal Effects											
Baseline Hurricane Effect				-0.0296** (0.014)	-0.0299** (0.014)		-0.0296** (0.014)	-0.0299** (0.014)	-0.0341** (0.015)	-0.0344** (0.015)	
Num. Cluster				-0.0220* (0.013)	-0.0222* (0.013)						
Increase 1											
Num. Branch											
Increase 1											
% Clustered Branch										-0.0256* (0.014)	-0.0257* (0.014)
Increase 10%											

Notes: The dependent variable in Panel A is firm-level branch-based aggregate sales growth. Columns (1)-(4) progressively introduce the moderation effect of the number of clusters across branches. Columns (5)-(7) introduce the moderation effect of the number of branches, and Columns (8)-(10) introduce the moderation effect of the share of clustered branches. Panel B reports marginal effects implied by the corresponding moderation specifications. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In Panel A, the estimated hurricane effect on firm-level sales growth is negative in all specifications, but it is statistically significant only in the moderation models. This pattern differs from the transaction-level results reported earlier. In Table 2, the dependent variable is supplier–customer transaction growth, which captures more direct and localized disruptions to individual trading relationships. There, hurricane exposure is associated with substantially larger declines in sales growth, on the order of 10 to 20 percentage points. By contrast, the branch-level analysis examines how a firm’s aggregate sales performance responds when part of its operational network is exposed to a hurricane. The smaller and less precisely estimated baseline coefficients in Table 4 are therefore consistent with firms’ ability to absorb or redistribute disruptions across branches.

The moderation results show that geographical organization plays an important role in shaping this aggregate resilience. The interaction between Hurricane and Number of Clusters is positive and statistically significant, indicating that firms with more spatially separated clusters experience smaller hurricane-related sales losses. This result remains robust after adding firm-size controls in Column (4), suggesting that the moderating role of clusters is not simply driven by larger firms having more extensive branch networks. A similar pattern appears for the number of branches: the interaction between Hurricane and Number of Branches is positive and statistically significant, including in the specification with firm-size controls in Column (7). This suggests that a broader branch network helps firms buffer localized disruptions even after accounting for time-varying firm scale. The share of clustered branches also moderates hurricane-related losses. The interaction between Hurricane and Percentage of Clustered Branches remains positive and statistically significant in Column (10), indicating that firms with a larger share of branches located in clusters are better able to maintain aggregate performance when hurricanes occur.

Panel B summarizes the economic magnitude of these moderating effects. Focusing on the specifications with firm-size controls, the implied hurricane effect in the cluster specification decreases in magnitude from -2.99 percentage points to -2.22 percentage points when the number of clusters increases by one. In the branch-count specification, adding one branch reduces the implied hurricane effect from -2.99 percentage points to -2.93 percent-

age points, suggesting a smaller but still directionally consistent buffering effect. In the clustered-branch-share specification, a 10 percentage point increase in the share of clustered branches reduces the implied hurricane effect from -3.44 percentage points to -2.57 percentage points. These results indicate that branch-level geographical organization can meaningfully attenuate hurricane-related sales losses at the firm level, and that this pattern is not solely explained by firm size.

Robustness Checks

We perform a host of robustness checks to assess the consistency of our previous findings by using alternative measurement strategies. To support our identifying assumption that natural disasters are an unanticipated shock and that they are uncorrelated with other local trends, we run specifications where we include the *lead* frequency of natural disaster exposure and the *contemporaneous* frequency. The results from these models across disaster types are reported in Appendix Table A5. None of the lead coefficients across models are statistically significant, supporting our identifying assumption, and the coefficient on hurricanes is estimated to be of a similar magnitude and precision.

For the heatmap in Figure 5, rather than filtering the agricultural sector individually for each regression, we use an interaction between agricultural sector and disaster type (agricultural sector * disaster). This approach ensures that we have the same number of observations across disaster types, particularly for those with limited data, such as tsunamis or earthquakes. By doing this, we ensure that the same observations are used to estimate fixed effects and control variables across disaster and firm types.

From Appendix Figure A5, we observe a consistent pattern in the impact of hurricanes. Hurricanes consistently exhibit a significant negative effect on key sectors such as Milling/Baking, Misc Food, and Beverage. This robustness across different models suggests that the adverse impact of hurricanes is not dependent on the specific measurement approach or limitations in sample size. The negative coefficient for hurricanes in these sectors remains stable, confirming that hurricanes disrupt production and supply chains in a reliable and predictable manner. Therefore, hurricanes serve as an ideal case for exploring diversification strategies

as a means of mitigating disaster-related disruptions.

We also perform a series of robustness checks regarding specification and data aggregation choices in Appendix Table A6. In columns 1, we only include large hurricane disasters, defined as those causing property damage greater than \$1 million. This approach allows us to focus on more severe hurricanes and examine whether the moderation effect is consistent under extreme conditions. As expected, the coefficient on large hurricanes is larger in absolute value than that on all hurricane events. In column 2, we apply weights based on company sales to account for differences in company size. Finally, in column 3, we exclude observations from 2020 onwards to eliminate potential confounding effects related to COVID-19, which might have impacted sales growth and disaster responses in ways unrelated to hurricanes.

Appendix Tables A7 and A8 follow this same pattern of presentation and robustness tests for the different measures of diversification. Across all models, the baseline impacts of hurricanes and the mitigation from diversification remain qualitatively and quantitatively similar.

5 Discussion and Policy Implications

This paper provides evidence that natural disasters can disrupt downstream food supply chains through both supplier–customer relationships and firm operating networks. The results are consistent with the production-network literature showing that localized shocks can propagate beyond directly affected firms through input-output and supplier–customer linkages (Barrot and Sauvagnat, 2016; Boehm, Flaaen, and Pandalai-Nayar, 2019; Carvalho et al., 2021). Our findings indicate that hurricanes generate the most consistent and economically meaningful reductions in supplier–customer transaction growth among food firms. This pattern indicates that damaging, infrastructure-relevant disasters can affect food supply chains beyond farm production by disrupting transaction flows, local establishments, employment, and the operational geography of firms.

We show that the resilience of firms to hurricane shocks depends on the type of organizational structure involved. Customer diversification and international diversification appear to attenuate hurricane-related losses, whereas business diversification provides weaker evidence

of mitigation. This distinction is consistent with recent work emphasizing that resilience strategies are not costless and that diversification may involve trade-offs between risk reduction, coordination costs, and allocative efficiency (Grossman, Helpman, and Lhuillier, 2023; Elliott, Golub, and Leduc, 2022; Kopytov et al., 2024). The branch-level results also show that the spatial organization of operations matters. Firms with more geographically dispersed branch networks, more clusters, and a higher share of clustered branches experience smaller hurricane-related losses at the aggregate firm level. This does not imply that simply adding branches is always optimal; rather, it suggests that operational geography can provide a buffer when firms have the capacity to reallocate activity across locations.

These findings have several policy implications. First, disaster preparedness for food systems should extend beyond farms and primary production. Public agencies and industry groups often focus on crop losses, livestock losses, and immediate agricultural damages, but our results suggest that downstream processors, wholesalers, warehouses, and distribution facilities are also important nodes of food-system resilience. Disaster planning should therefore include maps of food-industry establishments, major distribution corridors, cold-storage facilities, processing plants, and wholesale hubs, especially in hurricane-prone regions.

Second, promoting different forms of diversification may be one way to increase resilience to natural disasters. Expanding customer bases and stimulating regionalized capacity are proven methods to mitigate the economic losses. However, such interventions may not always be socially optimal, given that they may lead to allocative inefficiency and increase costs during normal periods. Future work can better balance the resilience benefits from diversification strategies versus the trade-offs to inform optimal policymaking.

Several limitations should be noted. First, the analysis focuses on U.S. firms and U.S. county-level disasters. The results therefore speak most directly to the domestic U.S. food-sector setting and should not be interpreted as a complete assessment of global food-system climate risk. Second, hurricanes receive particular attention in our analysis because they generate the clearest and most economically meaningful effects in our U.S. food-firm data. This does not imply that hurricanes are the most important hazard for all food systems or all regions.

Furthermore, some disaster effects may be imprecisely estimated because highly localized or short-duration shocks may not be fully captured by annual or quarterly firm outcomes or headquarters-based exposure measures. Future research could extend this analysis by using higher-frequency logistics data, facility-level production data, crop-loss measures, drought indices, or international firm networks to study other types of food-system shocks.

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Appendix: Clustering Methodology (HDBSCAN)

For geographical diversification, we introduce the HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) method, a robust clustering technique well-suited for identifying clusters in datasets with variable density and noise.

This method begins by calculating a “core distance” for each data point:

$$d_{core}(x_i) = distance(x_i, x_{k-th}) \quad (6)$$

This measure is defined as the distance to its k -th nearest neighbor, where k is a user-defined parameter that sets the minimum number of points required to consider a region as dense. In order to account for varying densities, HDBSCAN calculates “mutual reachability distance” between any two points. The mutual reachability distance between two points x_i and x_j is defined as the maximum of the core distances of each point and the actual distance between them.

$$d_{reach}(x_i, x_j) = max(d_{core}(x_i), d_{core}(x_j), d(x_i, x_j)) \quad (7)$$

This adjustment prevents loosely connected points in sparse areas from being included in clusters, ensuring that clusters are formed only in genuinely dense regions. Using these mutual reachability distances, HDBSCAN constructs a minimum spanning tree (MST), where each edge represents the mutual reachability distance between points. In Figure A3 (a), the blue lines that connect points within clusters, forming a network between them, represent the MST. This MST helps reveal the hierarchical structure of the data, as cutting the tree at various levels based on density thresholds allows for the detection of clusters at multiple densities. The hierarchical tree is then condensed by identifying clusters that persist over a range of density thresholds, reflecting their stability. Clusters that exhibit high stability are retained, while those that do not persist are discarded as noise.

Appendix: Supplemental Figures and Tables

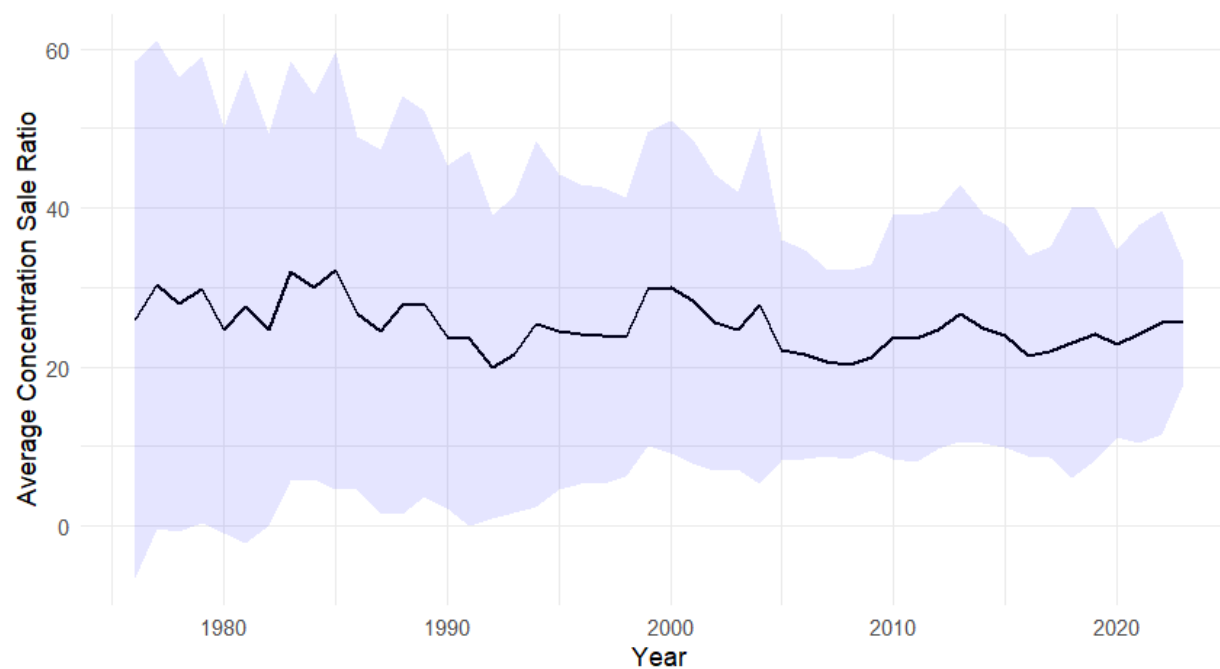


Figure A1: Trends in Sales Concentration

Note: The figure shows annual trends in customer sales concentration for firms in the sample over 1976–2023. The solid line reports the yearly mean sales concentration ratio, and the shaded area spans one standard deviation above and below the annual mean.

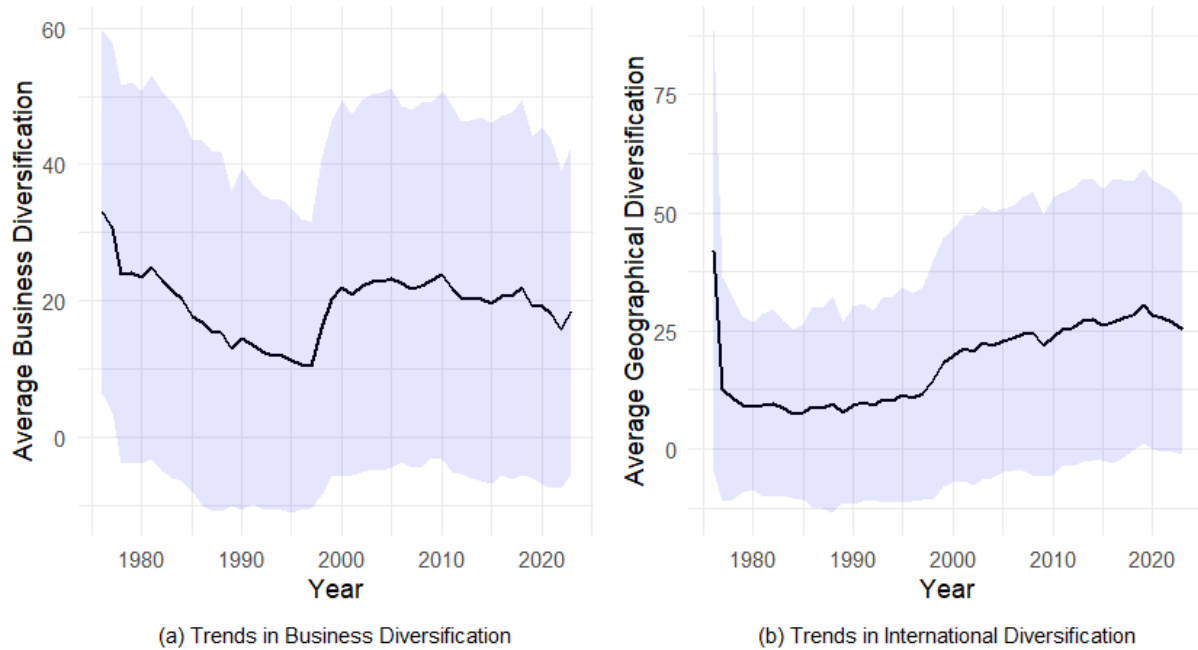
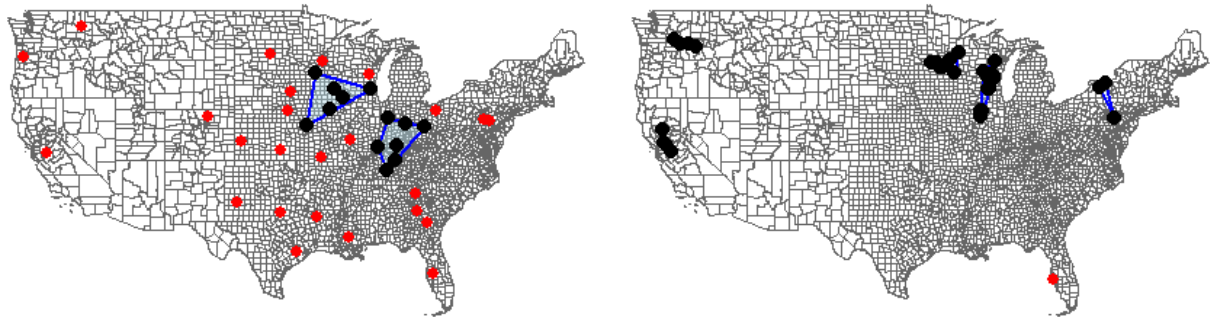


Figure A2: Trends in Business and International Geographical Diversification

Note: The figure reports annual trends in diversification for sample firms over 1976–2023. Panel (a) shows sales diversification across business lines, and panel (b) shows international sales diversification across geographical markets. In each panel, the solid line reports the yearly mean diversification index, and the shaded area spans one standard deviation above and below the annual mean.



(a) Loosely clustered branches

(b) Tightly clustered branches

Figure A3: Comparison of Different Cluster Scenarios

Note: The figure illustrates two stylized branch-network configurations for two firms. Panel (a) shows a firm with branches that are more loosely distributed across the space, while panel (b) shows a firm with branches that are more tightly concentrated. Red dots denote individual branch locations. Black dots denote branches that fall within the same cluster, with blue shaded regions highlighting the clustered area.

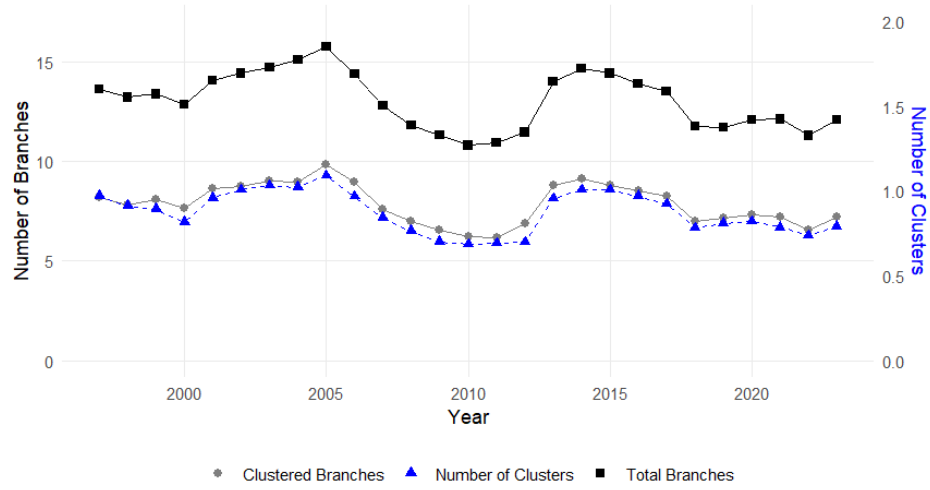


Figure A4: Trends in Branches and Clusters

Note: The figure reports annual trends in firms' branch-network structure over the period. The black squares (left axis) show the yearly mean number of total branches. The grey circles (left axis) show the yearly mean number of clustered branches. The blue triangles (right axis) show the yearly mean number of spatial clusters.

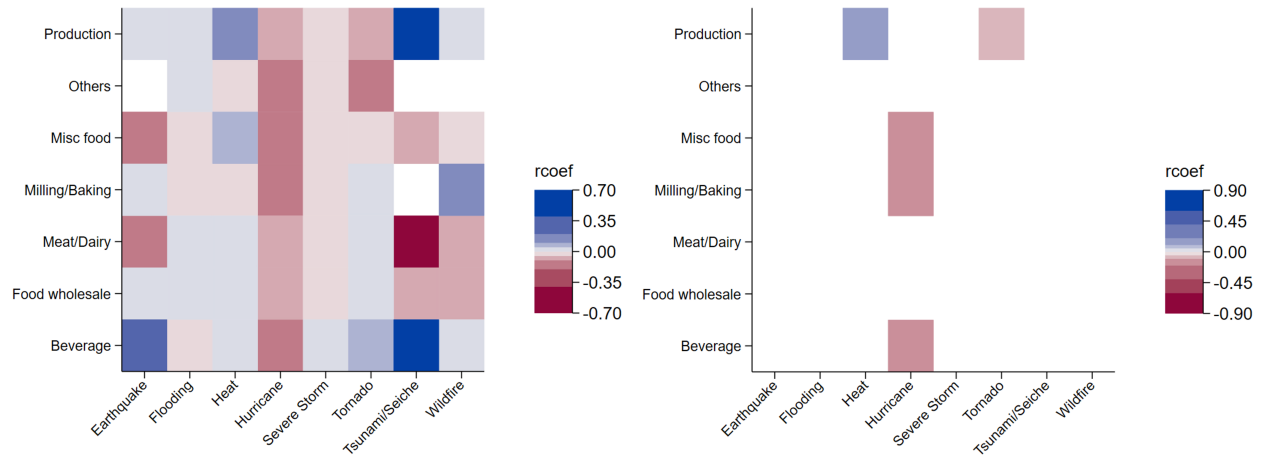


Figure A5: Robustness Check: Estimated Disaster Effects across Food Sectors

Note: Robustness check using a pooled regression with sector-by-disaster interactions, rather than separate regressions by sector. Right panel reports only statistically significant at the 5% level coefficients. Colors range from blue for positive effects to red for negative effects.

Tables

Table A1: Descriptive Statistics

Panel A: Firm Sales						
	Obs.	Mean	Std.Dev	p1	p50	p99
Sales growth	9,197	0.143	0.569	-0.828	0.056	0.882
Num. customers	9,197	3.449	3.878	1	2	21
Hurricane shock	9,197	0.056	0.263	0	0	1
Large hurricane	9,197	0.021	0.148	0	0	1
Flooding shock	9,197	0.794	1.178	0	0	5
Large flooding	9,197	0.159	0.432	0	0	2
Concentration sale ratio	3,843	24.538	15.304	2.071	20.60	80.086
Concentration sale ratio growth	3,625	10.152	115.43	-68.64	0	209.11
BUSSEG diversification index	8,399	18.823	25.092	0	0	77.549
GEOSEG diversification index	7,625	19.875	25.790	0	1.544	78.410
Food Company Financial Information:						
ROA	10,595	-0.465	12.226	-6.356	0.040	0.351
Size	10,648	5.187	2.886	-2.172	5.243	11.058
R&D to Sales	4,125	0.212	4.657	0	0.005	2.195
Gross Profit Margin	10,443	-0.083	10.682	-2.333	0.290	0.786
Operating Leverage	10,177	2.005	5.644	0.109	1.305	9.827
Cash Holdings	10,642	0.109	0.157	0	0.050	0.800
Panel B: Establishment Data						
Change in establishment count	66,255	1.262	8.616	-10	0	28
Change in employment level	66,255	7.626	324.780	-521	1	613
e.g. Hurricane shock	68,390	0.042	0.224	0	0	1
e.g. Large hurricane	68,390	0.014	0.127	0	0	1
e.g. Flooding shock	68,390	0.439	0.853	0	0	4
e.g. Large flooding	68,390	0.053	0.245	0	0	1
Panel C: Branch Data						
Sales growth	20,397	-0.060	0.369	-0.986	0	0.982
Num. clusters	20,103	1.166	5.314	0	0	21
Num. branches	20,103	16.826	67.803	1	3	247
Clustered branches percentage	20,103	15.166	32.367	0	0	100
Hurricane shock	20,397	0.045	0.233	0	0	1
Large hurricane	20,397	0.018	0.139	0	0	1

Notes: This table reports descriptive statistics for the study samples. Panel A summarizes firm sales, customer relationships, disaster exposure, sales concentration, diversification, and financial characteristics. Panel B summarizes annual changes in county-level establishment and employment counts and county-level disaster exposure. Panel C summarizes parent companies' sales growth, branches, clusters, and disaster exposure. Obs. reports the number of nonmissing observations; p1, p50, and p99 denote the 1st, 50th, and 99th percentiles, respectively.

Table A2: Frequency and Property Damage of Natural Disasters by Disaster Type, 1976–2023

Disaster Type	Total frequency	Total property damage, 2023 USD	Share of total frequency	Share of total damage
Tornado	28,090	\$63.06 billion	4.2%	6.7%
Hurricane	6,277	\$287.98 billion	0.9%	30.6%
Flooding	70,629	\$297.68 billion	10.5%	31.6%
Earthquake	71	\$59.64 billion	0.0%	6.3%
Tsunami	60	\$0.08 billion	0.0%	0.0%
Thunderstorm	166,669	\$29.76 billion	24.8%	3.2%
Heat	7,636	\$0.59 billion	1.1%	0.1%
Other hazards	392,208	\$201.89 billion	58.4%	21.5%
All hazards	671,640	\$940.69 billion	100.0%	100.0%

Notes: This table summarizes natural disaster exposure in the United States over 1976–2023. It reports, for each disaster type, the total number of recorded disaster events, total property damage adjusted to 2023 dollars, and the corresponding shares of total disaster frequency and total property damage.

Table A3: SIC codes used to define food-sector firms

SIC	Title	SIC	Title
0100	Agricultural Production Crops	0110	Cash Grains
0111	Wheat	0112	Rice
0115	Corn	0116	Soybeans
0119	Cash Grains, Not Elsewhere Classified	0130	Field Crops, Except Cash Grains
0131	Cotton	0132	Tobacco
0133	Sugarcane and Sugar Beets	0134	Irish Potatoes
0139	Field Crops, Except Cash Grains, Not Elsewhere Classified	0160	Vegetables and Melons
0161	Vegetables and Melons	0170	Fruits and Tree Nuts
0171	Berry Crops	0172	Grapes
0173	Tree Nuts	0174	Citrus Fruits
0175	Deciduous Tree Fruits	0179	Fruits and Tree Nuts, Not Elsewhere Classified
0180	Horticultural Specialties	0181	Ornamental Floriculture and Nursery Products
0182	Food Crops Grown Under Cover	0190	General Farms, Primarily Crop
0191	General Farms, Primarily Crop		
0200	Agricultural Production Livestock and Animal Specialties	0210	Livestock, Except Dairy and Poultry
0211	Beef Cattle Feedlots	0212	Beef Cattle, Except Feedlots
0213	Hogs	0214	Sheep and Goats
0219	General Livestock, Except Dairy and Poultry	0240	Dairy Farms
0241	Dairy Farms	0250	Poultry and Eggs
0251	Broiler, Fryer, and Roaster Chickens	0252	Chicken Eggs
0253	Turkeys and Turkey Eggs	0254	Poultry Hatcheries
0259	Poultry and Eggs, Not Elsewhere Classified	0270	Animal Specialties
0271	Fur-bearing Animals and Rabbits	0272	Horses and Other Equines
0273	Animal Aquaculture	0279	Animal Specialties, Not Elsewhere Classified
0290	General Farms, Primarily Livestock and Animal Specialties	0291	General Farms, Primarily Livestock and Animal Specialties
0700	Agricultural Services	0710	Soil Preparation Services
0711	Soil Preparation Services	0720	Crop Services
0721	Crop Planting, Cultivating, and Protecting	0722	Crop Harvesting, Primarily By Machine
0723	Crop Preparation Services For Market, Except Cotton Ginning	0724	Cotton Ginning
0740	Veterinary Services	0741	Veterinary Services For Livestock
0742	Veterinary Services For Animal Specialties	0750	Animal Services, Except Veterinary
0751	Livestock Services, Except Veterinary	0752	Animal Specialty Services, Except Veterinary
0760	Farm Labor and Management Services	0761	Farm Labor Contractors and Crew Leaders
0762	Farm Management Services	0780	Landscape and Horticultural Services
0782	Lawn and Garden Services	0783	Ornamental Shrub and Tree Services
0910	Commercial Fishing	0912	Finfish
0913	Shellfish	0919	Miscellaneous Marine Products
2000	Food and Kindred Products	2010	Meat Products
2011	Meat Packing Plants	2013	Sausages and Other Prepared Meat Products
2015	Poultry Slaughtering and Processing	2020	Dairy Products
2021	Creamery Butter	2022	Natural, Processed, and Imitation Cheese
2023	Dry, Condensed, and Evaporated Dairy Products	2024	Ice Cream and Frozen Desserts
2026	Fluid Milk	2030	Canned, Frozen, and Preserved Fruits, Vegetables, and Food Specialties

SIC Title	SIC Title
2032 Canned Specialties	2033 Canned Fruits, Vegetables, Preserves, Jams, and Jellies
2034 Dried and Dehydrated Fruits, Vegetables, and Soup Mixes	2035 Pickled Fruits and Vegetables, Vegetable Sauces and Seasonings, and Salad Dressings
2037 Frozen Fruits, Fruit Juices, and Vegetables	2038 Frozen Specialties, Not Elsewhere Classified
2040 Grain Mill Products	2041 Flour and Other Grain Mill Products
2043 Cereal Breakfast Foods	2044 Rice Milling
2045 Prepared Flour Mixes and Doughs	2046 Wet Corn Milling
2047 Dog and Cat Food	2048 Prepared Feeds and Feed Ingredients for Animals and Fowls, Except Dogs and Cats
2050 Bakery Products	2051 Bread and Other Bakery Products, Except Cookies and Crackers
2052 Cookies and Crackers	2053 Frozen Bakery Products, Except Bread
2060 Sugar and Confectionery Products	2061 Cane Sugar, Except Refining
2062 Cane Sugar Refining	2063 Beet Sugar
2064 Candy and Other Confectionery Products	2066 Chocolate and Cocoa Products
2067 Chewing Gum	2068 Salted and Roasted Nuts and Seeds
2070 Fats and Oils	2074 Cottonseed Oil Mills
2075 Soybean Oil Mills	2076 Vegetable Oil Mills, Except Corn, Cottonseed, and Soybean
2077 Animal and Marine Fats and Oils	2079 Shortening, Table Oils, Margarine, and Other Edible Fats and Oils
2080 Beverages	2082 Malt Beverages
2083 Malt	2084 Wines, Brandy, and Brandy Spirits
2085 Distilled and Blended Liquors	2086 Bottled and Canned Soft Drinks and Carbonated Waters
2087 Flavoring Extracts and Flavoring Syrups, Not Elsewhere Classified	2090 Miscellaneous Food Preparations and Kindred Products
2091 Canned and Cured Fish and Seafoods	2092 Prepared Fresh or Frozen Fish and Seafoods
2095 Roasted Coffee	2096 Potato Chips, Corn Chips, and Similar Snacks
2097 Manufactured Ice	2098 Macaroni, Spaghetti, Vermicelli, and Noodles
2099 Food Preparations, Not Elsewhere Classified	
2100 Tobacco Products	2110 Cigarettes
2111 Cigarettes	2120 Cigars
2121 Cigars	2130 Chewing and Smoking Tobacco and Snuff
2131 Chewing and Smoking Tobacco and Snuff	2140 Tobacco Stemming and Redrying
2141 Tobacco Stemming and Redrying	
5140 Groceries and Related Products	5141 Groceries, General Line
5142 Packaged Frozen Foods	5143 Dairy Products, Except Dried or Canned
5144 Poultry and Poultry Products	5145 Confectionery
5146 Fish and Seafoods	5147 Meats and Meat Products
5148 Fresh Fruits and Vegetables	5149 Groceries and Related Products, Not Elsewhere Classified
5150 Farm-Product Raw Materials	5153 Grain and Field Beans
5154 Livestock	5159 Farm-Product Raw Materials, Not Elsewhere Classified
5180 Beer, Wine, and Distilled Alcoholic Beverages	5181 Beer and Ale
5182 Wine and Distilled Alcoholic Beverages	

Table A4: Effects of Major Disasters on Non-Food Firms' Sales Growth

Sales Growth (t-4, t)							
	Tornado	Hurricane	Flooding	Earthquake	Tsunami	Thunderstorm	Heat
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Disaster	-0.018	-0.071	-0.091**	0.047	0.092	0.020	-0.026
	(0.029)	(0.061)	(0.039)	(0.086)	(0.078)	(0.016)	(0.080)
Supplier-Customer FE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Obs.	70,842	70,842	70,842	70,842	70,842	70,842	70,842
R^2	0.361	0.361	0.361	0.361	0.361	0.361	0.361

Notes: The table shows estimated coefficients from regressions of non-food-company supplier-customer transaction-level sales growth on exposure to natural disasters. Columns (1)–(7) use tornado, hurricane, flooding, earthquake, tsunami, severe thunderstorm, and heat as the main independent variables, respectively. Regressions include supplier-customer pair fixed effects and year fixed effects. Standard errors are clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Current and Lead Effects of Natural Disasters on Food Firm Sales Growth

	Sales Growth (t-4, t)						
	Tornado	Hurricane	Flooding	Earthquake	Tsunami	Thunderstorm	Heat
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Current-year Disaster Shock ($t = 0$)	0.018 (0.016)	-0.111*** (0.034)	-0.001 (0.008)	-0.101 (0.069)	-0.116 (0.141)	-0.002 (0.007)	0.025** (0.011)
One-year Lead Disaster Shock ($t = +1$)	0.016 (0.018)	-0.006 (0.042)	0.001 (0.01)	0.017 (0.077)	0.002 (0.107)	-0.001 (0.007)	-0.019 (0.013)
Supplier-Customer FE	YES	YES	YES	YES	YES	YES	YES
Region-Year FE	YES	YES	YES	YES	YES	YES	YES
Obs.	7,866	7,866	7,866	7,866	7,866	7,866	7,866
R^2	0.285	0.287	0.285	0.285	0.285	0.285	0.285

Notes: The table reports estimated coefficients from regressions of food firm supplier-customer-level sales growth on current-year disaster exposure and a one-year lead of disaster exposure. Each column corresponds to a different disaster type. All regressions include supplier-customer pair fixed effects and region-year fixed effects. Standard errors are clustered at the county level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Robustness Checks for the Moderating Effect of Changes in Sales Concentration

	Sales Growth (t-4, t)					
	Large Hurricanes		Weighting		Exclude COVID	
	(1)	(1)	(2)	(2)	(3)	(3)
Hurricane	-0.1530*** (0.049)	-0.1342*** (0.035)	-0.0674*** (0.018)	-0.0283 (0.034)	-0.1155*** (0.031)	-0.0927** (0.045)
Concentration Sale Growth		0.0006*** (0.000)		0.0006*** (0.000)		0.0006*** (0.000)
Hurricane *		-0.0009*** (0.000)		0.0021 (0.003)		-0.0005** (0.000)
Supplier-Customer FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	8,529	3,341	8,529	3,341	7,073	2,862
R-Squared	0.253	0.339	0.379	0.303	0.259	0.343

Notes: Column (1) uses an alternative measure of disaster intensity. Column (2) reports estimates weighted by firm size. Column (3) excludes the effect from the COVID-19 period. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Robustness Checks for the Moderating Effects of Segment Diversification

	Sales Growth (t-4, t)					
	Large Hurricanes		Weighting		Exclude COVID	
	(1)	(1)	(2)	(2)	(3)	(3)
Hurricane	-0.1543** (0.065)	-0.1898*** (0.045)	-0.0727** (0.030)	-0.0592* (0.031)	-0.1118*** (0.040)	-0.2051*** (0.047)
Business Diversification	-0.0015* (0.001)		0.0019*** (0.000)		-0.0013 (0.001)	
Hurricane * Business Diversification	-0.0001 (0.002)		4e-5 (0.001)		-0.0004 (0.001)	
International Diversification		0.0017* (0.001)		-0.0002 (0.000)		0.0018* (0.001)
Hurricane * Int'l Diversification		0.0038** (0.002)		-0.0006 (0.001)		0.0035** (0.002)
Supplier-Customer FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Observations	7,770	7,083	7,770	7,083	6,465	5,964
R-Squared	0.259	0.249	0.418	0.383	0.265	0.251

Notes: Column (1) uses an alternative measure of disaster intensity. Column (2) reports estimates weighted by firm size. Column (3) excludes the effect from the COVID-19 period. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Robustness Checks for the Moderating Effects of Branch Network Structure

	Sales Growth (t-4, t)								
	Large Hurricanes			Weighting			Exclude COVID		
	(1)	(1)	(1)	(2)	(2)	(2)	(3)	(3)	(3)
Hurricane	-0.0236*	-0.0211*	-0.0218*	-0.0039	-0.0011	-0.0137*	-0.0204**	-0.0207**	-0.0191**
	(0.0137)	(0.0128)	(0.0132)	(0.006)	(0.006)	(0.007)	(0.009)	(0.008)	(0.009)
Num. Cluster	0.0016***			0.0009***			0.0018***		
	(5e-4)			(0.000)			(6e-4)		
Hurricane *	0.0081***			0.0083***			0.0079***		
Num. Cluster	(0.003)			(0.001)			(0.003)		
Num. Branch		0.0001***			0.0001***			0.0002***	
		(4e-5)			(1e-5)			(4e-5)	
Hurricane *		0.0004**			0.0005***			0.0006***	
Num. Branch		(2e-4)			(4e-5)			(2e-4)	
Percentage of			0.0006***			0.0004***			0.0006***
Clustered Branch			(9e-5)			(6e-5)			(1e-4)
Hurricane * %			-0.0002			0.0003**			0.0004
Clustered Branch			(4e-4)			(1e-4)			(2e-4)
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
Observations	19,986	19,986	19,986	19,950	19,950	19,950	14,696	14,696	14,696
R-Squared	0.181	0.181	0.183	0.180	0.179	0.174	0.193	0.194	0.195

Notes: Column (1) uses an alternative measure of disaster intensity. Column (2) reports estimates weighted by firm size. Column (3) excludes the effect from the COVID-19 period. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.